Multimodal data and fusion

MULTIMODAL DATA

Medical image



pixels

Radiology report

FINDINGS
Lungs are clear. Heart size is normal. No effusion or pneumothorax.

IMPRESSION
No acute cardiopulmonary disease.

free text

EHR

Age, sex	54, F
Temp	37.8 °C
WBC	11.2 $\times 10^9$/L
History	asthma
ICD-10	J45.909

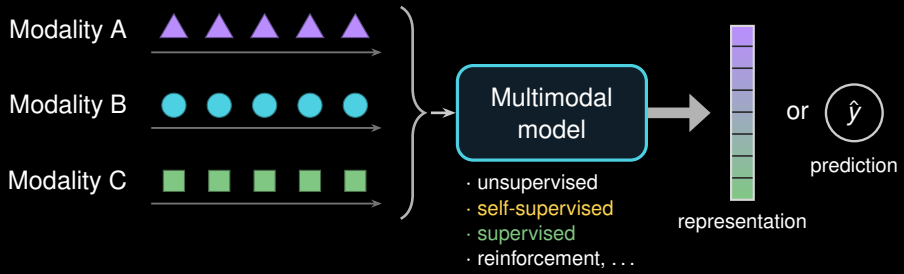
numbers and codes

one patient

Biobanks, health records, imaging and wearable sensors now set the stage for **multimodal AI** that captures the **complexity of human health and disease**.

Acosta et al., *Multimodal biomedical AI*, Nature Medicine

MULTIMODAL CLINICAL AI

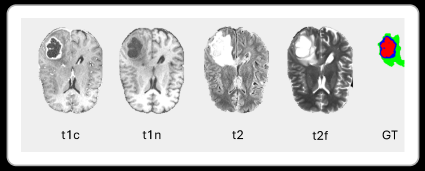


Morency & Liang, *Multimodal Machine Learning*, CMU 11-777

More on multimodal data

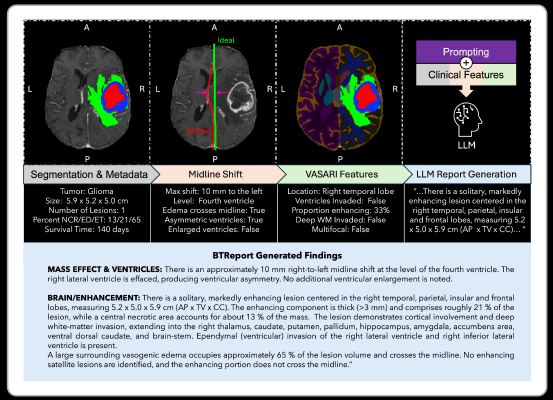
Special case

Nearly identical MRIs from tumor segmentation tasks.



Common case

MRI and radiology report.

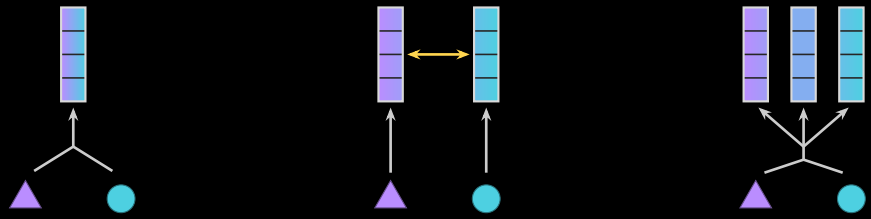


Ren et al., *Here Comes the Explanation: A Shapley Perspective on Multi-contrast Medical Image Segmentation*, arXiv; Heras Rivera et al., *BTReport: A Framework for Brain Tumor Radiology Report Generation with Clinically Relevant Features*, MIDL

THREE DIFFERENT STRATEGIES IN MULTIMODAL LEARNING

Representation: learning representations that reflect cross-modal interactions between individual elements, across different modalities.

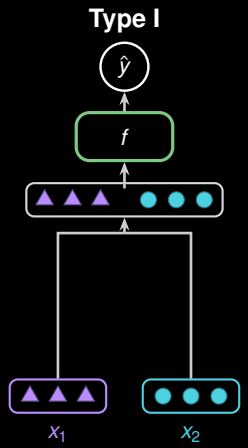
Fusion **Coordination** **Fission**

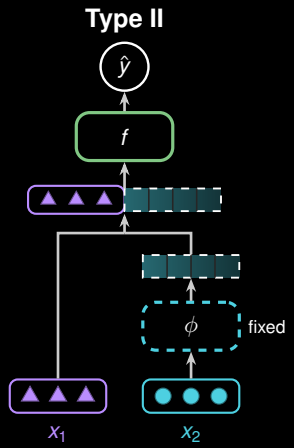


The rest of today is about fusion, and one question: **at what depth** in the network do the modalities meet?

Early, joint and late fusion

EARLY FUSION





$\hat{y} = f([x_1 ; x_2])$ $\min_f \mathcal{L}$

Original inputs, concatenated.

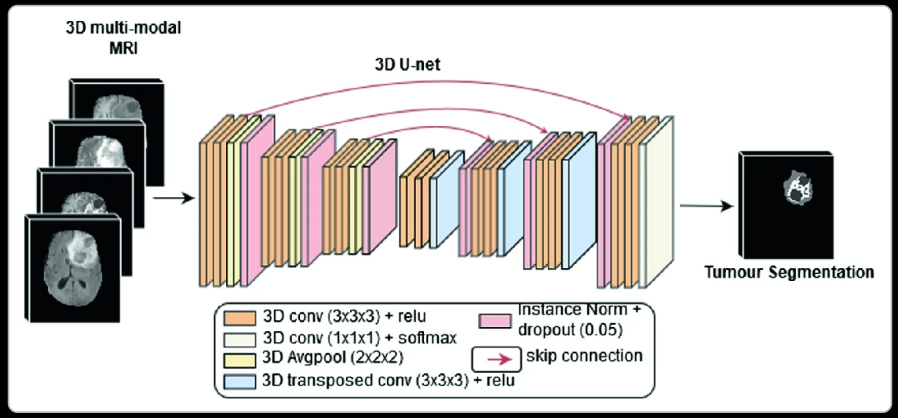
$\hat{y} = f([x_1 ; \phi(x_2)])$ $\min_f \mathcal{L}$

One input pre-extracted by a fixed ϕ.

Fuse before the model; the loss trains only f.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

EARLY FUSION: STACK THE MODALITIES AS CHANNELS



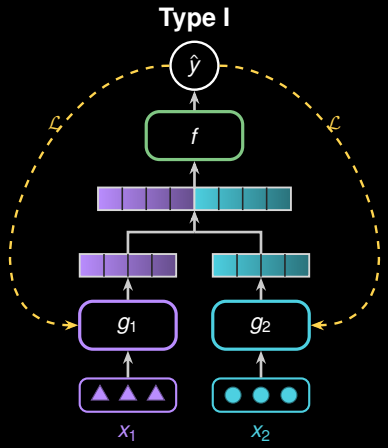
To segment brain tumours, the four 3D MRI sequences enter a 3D U-Net together as its input channels, so the very first convolution already fuses them (**early fusion, Type I**).

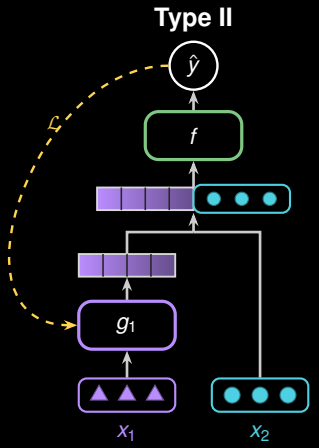
Mehta & Arbel, *3D U-Net for Brain Tumour Segmentation*, MICCAI Brainlesion Workshop

Limitations of input-level stacking

- **Incompatible spatial data.** A 3D MRI volume and a 2D X-ray projection differ in dimensionality and spatial correspondence, preventing direct channel-wise stacking.
- **Heterogeneous data types.** Clinical notes, laboratory measurements, and medication records lack the spatial structure required for direct stacking with image inputs.
- **Missing modalities at inference.** Models trained on a fixed set of input channels generally require all corresponding modalities at inference.
- **Alternative: joint fusion.** Modality-specific encoders map each input to a feature representation, which can then be combined for downstream prediction. Robustness to missing modalities requires explicit architectural or training strategies.

JOINT FUSION





$$\hat{y} = f([\,g_1(x_1)\,;\,g_2(x_2)\,]) \quad \min_{f,\,g_1,\,g_2} \mathcal{L}$$

Every modality gets a trained encoder.

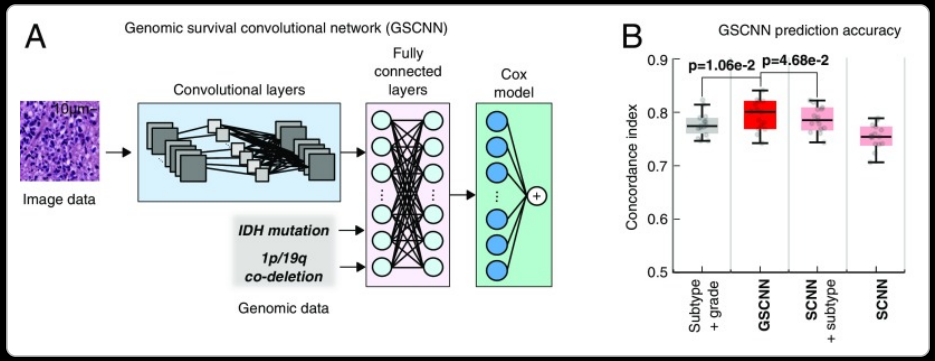
$$\hat{y} = f([\,g_1(x_1)\,;\,x_2\,]) \quad \min_{f,\,g_1} \mathcal{L}$$

One encoder; the other input joins raw.

Fuse learned **features**; the loss reaches every encoder.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

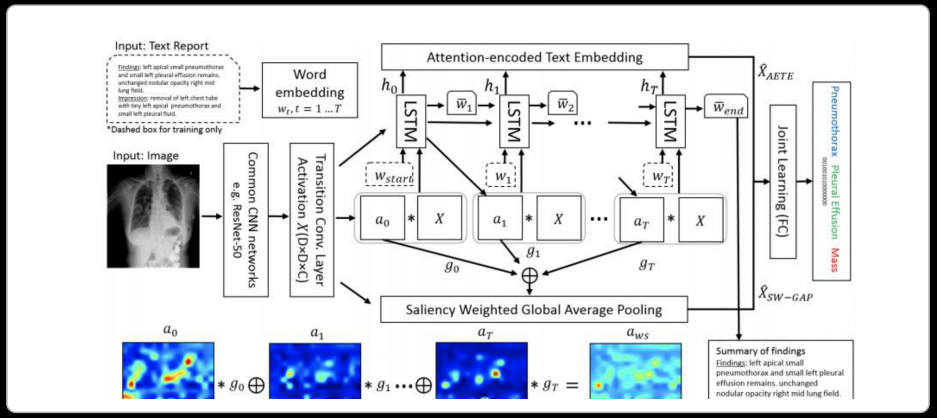
JOINT FUSION: HISTOLOGY AND GENOMICS



For glioma survival, IDH and 1p/19q status enter the CNN's fully connected layers and one Cox loss trains the whole network (**joint fusion, Type II**), raising the median c-index from 0.754 (histology alone) to **0.801**.

Mobadersany et al., *Predicting cancer outcomes from histology and genomics using convolutional networks*, PNAS

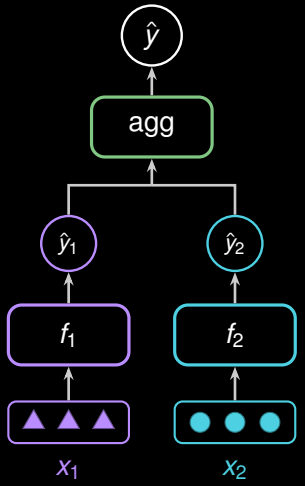
IMAGE AND TEXT, FUSED WITH ATTENTION



At **every word** of the report, a recurrent network decides which x-ray regions to look at. That is **attention**: a learned weighted average of the CNN's image features, with weights that sum to one. The fused representation predicts disease labels, which then serve as **automatic annotations** for x-rays with a report but no label.

Wang et al., *TieNet: Text-Image Embedding Network for Common Thorax Disease Classification and Reporting in Chest X-rays*, CVPR

Late fusion



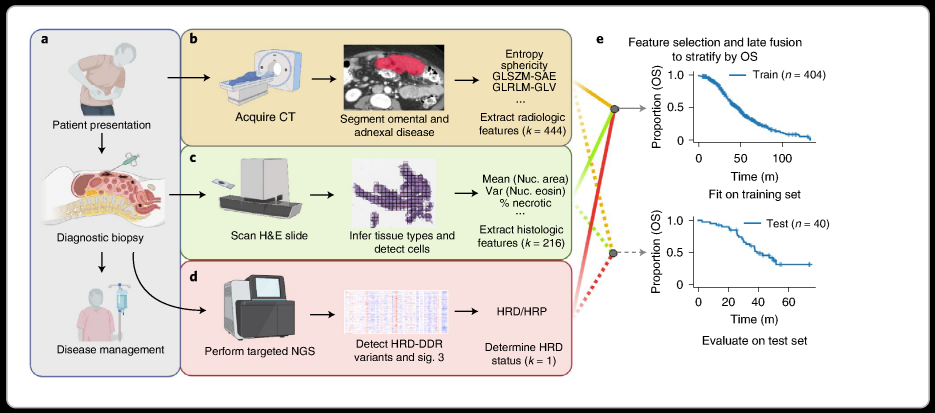
$$\hat{y} = \mathrm{agg}\big(f_1(x_1), f_2(x_2)\big)$$

$$\min_{f_1} \mathcal{L}(\hat{y}_1, y) \qquad \min_{f_2} \mathcal{L}(\hat{y}_2, y)$$

Fuse **predictions**. Each modality is its own model, trained alone; agg is a fixed rule such as an average, or a small trained model.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

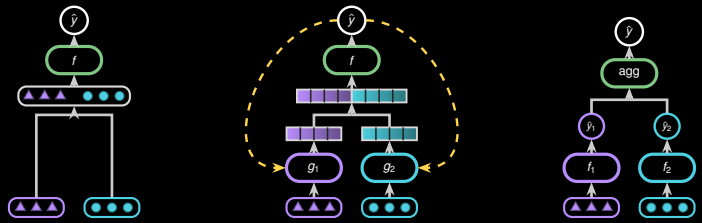
LATE FUSION: CT, HISTOLOGY AND GENOMICS



For ovarian cancer survival, each modality gets its own Cox model and a final Cox model fuses their risk scores. The best test model used **CT and histology alone** (c-index 0.62); adding all four modalities did **worse**. More modalities are no guarantee.

Boehm et al., *Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer*, Nature Cancer

EARLY, JOINT OR LATE: CHOOSING A DEPTH



	Early	**Joint**	**Late**
fuses	raw inputs	learned features	predictions
the loss trains	*f* only	*f* and every encoder	each model alone
choose it when	inputs share a grid, and all are present	each modality should shape the other's features	a modality may be missing: each model runs on its own

And whichever depth you pick: **more modalities are no guarantee**. Boehm's best model used two of its four.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine; Boehm et al., *Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer*, Nature Cancer